AI Shopping Agent

Project Report | Database Systems

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Introduction

AI Shopping Agent is a smart desktop application that helps users search and compare product prices across multiple Pakistani e-commerce platforms simultaneously. Users enter a product name, and the system retrieves relevant listings with prices and purchase links from Daraz, Goto, and Shophive. All search history, products, and user favorites are persisted in a MongoDB database, enabling intelligent caching and data-driven insights.

Objectives

- Make online product searching fast and easy for Pakistani consumers.
- Enable real-time price comparison across 3 major e-commerce sites.
- Persist user data (searches, favorites, products) using MongoDB.
- Provide analytics — best deals, price trends, and website stats.
- Use parallel scraping (threading) for optimal performance.

Technologies Used

Python 3.x	Core application language
MongoDB	NoSQL database (non-relational)
PyMongo	Python-MongoDB driver
Selenium	Dynamic web scraping
CustomTkinter	Modern GUI framework
Threading	Parallel scraping (3 sites)

Key Features

- **Product Search:** Search across Daraz, Goto & Shophive simultaneously
- **Price Comparison:** Side-by-side price comparison with best deal highlight
- **Favorites:** Save and manage favorite products persistently
- **Search History:** Full history with deduplication and caching
- **Analytics:** Website stats, savings comparisons, category insights
- **GUI Interface:** Modern desktop UI built with CustomTkinter

Database Design (MongoDB)

Collection	Fields
`all_products`	name, price, price_display, website, category, link, saved_at
`search_history`	product, date, time
`favorite_products`	product, price, website, link, date

MongoDB operations include full CRUD, aggregation pipelines for analytics, regex-based search, and duplicate-prevention logic via upsert patterns.

Conclusion

AI Shopping Agent successfully demonstrates the integration of web scraping, NoSQL database management, and GUI development. The project applies MongoDB collections and aggregation pipelines as a real-world non-relational database backend. Parallel scraping with Python threading ensures fast results, while intelligent caching reduces redundant requests. The application provides genuine value to Pakistani online shoppers by centralizing price comparisons across Daraz, Goto, and Shophive in a single desktop interface.